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SYSTEMATIC REVIEW

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Physical activity wishes of people with excess body weight - a qualitative systematic review

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Abstract

Background Despite significant public health focus upon minimizing overweight and obesity, the prevalence of individuals with such conditions continues to rise globally. This involves major health challenges for the individual, as well as increased global healthcare costs. Diet and physical activity are known to be an effective method of weight loss. However, research also indicates that it is often difficult for individuals to maintain increased levels of physical activity. Thus it is important to investigate the wishes of people with overweight and obesity to understand how to best support maintaining adequate levels of physical activity. Many articles have been published in the field, but to our knowledge no systematic review has explored wishes of people with overweight and obesity for physical activity. The aim of this qualitative systematic review was to identify how physical activity should be planned and structured to include the wishes and needs of people with excess body weight to increase their level of physical activity.

Materials and methods Systematic review guidelines have been followed. A systematic literature search has been carried out in the following databases: PubMed, CINAHL, MEDLINE, EMBASE, PsycINFO and Cochrane. Data synthesis has been conducted according to Thomas and Harden's thematic analysis.

Results The search resulted in 15,333 articles after removal of duplicates, with 10 articles included in this qualitave systematic review. The included studies were carried out in different countries and cultures. Two primary themes, coping and environment, and seven sub themes emerged from this analysis. Each theme encompassed specific barriers and facilitators that shape the ability of individuals with excess body weight to engage in physical activity.

Conclusions None of the found studies explicitly examined the wishes of individuals with excess body weight regarding physical activity. However, the studies do explore various perspectives on physical activity from the viewpoint of individuals with overweight and obesity. These insights contribute valuable knowledge to the planning and organization of physical activity, particularly in terms of its form and content.

Keywords Obesity, Overweight, Excess body weight, Wishes, Needs, Physical activity

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Introduction

The prevalence of overweight and obesity are increasing worldwide to an extent that is considered a global epidemic [1, 2]. In Europe, over 50% of the population is either overweight or obese [3]. This is a public health problem with rising hospitalizations and increased mortality [4, 5]. Both of these conditions are correlated with an increased risk of diabetes, hypertension, coronary heart diseases, and different kinds of cancer [4–6]. Therefore, the World Health Organization (WHO) has aimed to reduce the rise in overweight and obesity [1]. WHO defines overweight as having a BMI ≥ 25 and obesity as having a BMI ≥ 30 [7]. Throughout this paper, excess body weight (EBW) is used to refer to people with overweight or obesity collectively.

A combination of diet and increased physical activity are known to contribute to weight loss [8]. Moreover, increased physical activity has beneficial effects on overall health, even if weight loss is not achieved [8]. The Danish National Board of Health stated in a 2022 report that over half of people with EBW want to change their health behaviors, including increasing physical activity [9]. Several studies describe barriers for people with EBW in maintaining or increasing their level of physical activity. Such barriers include that indviduals feel "too obese" to be physically active, that women with EBW find physical activity too difficult [10], that individuals do not identify themselves as "sporty" [11], and that women are prone to the affects of stigma causing them to avoid the gym [12]. A study by Rech et al. describes that it is difficult to maintain an increased level of physical activity, and that half of people who increase their physical activity will resume their prior level of physical activity after half a year [13]. The authors describe that this behavioral pattern is more common in people with EBW [13]. These findings are further supported by the WHO, which found that more than 27% of adults do not meet recommendations for physical activity [14].

Understanding that there are multiple barriers that can influence physical activity participation, it is important to consider how personal wishes can support such adherence. Physical activity wishes are the preferences of an individual regarding the details of how they engage in exercise [12]. Significant research considers motivation, or motives, for physical activity participation [15]. However, why an individual participates (motivation) is unique to how they desire to participate (wishes). Therefore, it is equally important to explore how individuals wish to participate in physical activity routines, as this knowledge can improve understanding about how to best structure interventions to support the preferences of individuals with EBW.

Despite several studies that have investigatedbarriers to physically activity faced by people with EBW [10, 11, 16], understanding the wishes of people with EBW regarding the content and form of increasing physical activity contains gaps in knowledge. Specifically, there has not been a previous review that has explored this aspect of physical activity for individuals with EBW. A systematic review was valuable to explore on this topic to help syntehsize evidence to help more appropriately guide physical activity interventions for individuals with EBW [17]. Therefore, the aim of this study was to identify how physical activity must be planned and organized in regards to form and content to encompass the wishes of people with EBW to support increasing their level of physical activity.

Methods

This qualitative systematic review has been registered in PROSPERO (CRD42023396098), and was conducted according to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines [16], and the principles of the Helsinki Declaration (2000) [18].

Inclusion and exclusion criteria

The Population, Phenomena of Interest, and Context (PICo) [19], were used to define the eligibility criteria for this systemtic review. Articles with the following criteria were included:

- Population: Studies that investigated adults (aged ≥ 18 years) with overweight or obesity (Body Mass Index (BMI) ≥ 25). Studies on pregnancy, critical illnesses such as cancer, neurological disorders such as Parkinson's or stroke, severe psychiatric disorders such as depression, schizophrenia, or psychosis, and patients having severe pain were excluded, as these groups have distinct considerations compared to the general adult population with excess body weight.
- Phenomena of Interest: Studies that investigated wishes, needs, and experiences with physical activity.
- Context: Studies that investigated physical activity and exercise.
- Study design: Qualitative studies that examined the wishes and needs of people with EBW in relation to physical activity. No restrictions were applied regarding intervention length. Only peer reviewed full-text articles that were available in English were included in this review.

Data sources and search strategy

In September 2024 a systematic literature search was conducted in the following online databases: PubMed, Cochrane, CINAHL, EMBASE, MEDLINE, and PsycINFO. The search strategy included subheadings (MeSH terms) and keywords. There was a register search in OpenGray and Google Scholar. Other references were searched through key articles. Two authors (CB and SSN) conducted the initial search. Search terms included "obesity", "needs", and "physical activity". The search was organized in three building blocks combined with the Boolean operator "AND". Within the main keyword groups, the Boolean operator "OR" was used to separate similar terms, allowing for increased sensitivity of the search. The search terms were adapted to each database. In PubMed and CINAHL, the subheadings used for the MeSH term obesity were "rehabilitation" and "therapy". In EMBASE, the subheading used for the MeSH term obesity was "obesity patient". All searches were limited to English.

Study selection

All searches were uploaded in the reference program Endnote 20 and duplicates were removed. All references were exported to XML format in batches database by database, and the batches were imported to the software program Covidence for screening. Records were screened for lack of relevance in their title and abstract by the first two authors independently. Any disagreements during study selection were resolved through discussion between the first two authors (JRC and CB) until consensus was reached.The articles were divided in four batches and the full texts were screened in each batch for eligibility by eight of the authors, who worked together two and two independently. The eight authors were included and thus grouped in pairs. A selection form with predefined inclusion criteria was used by the authors to determine elibility of the articles they reviewed. If one or more inclusion criteria was not met, the article was excluded, and the reason for exclusion was recorded. Discrepancies in one of the four groups was resolved by the first author (JRC).

Risk of bias assessment

Risk of bias was assessed by a minimum of two authors independently. The CASP Qualitative Studies Checklist was used (Critical Appraisals Skills Programme. CASP Checklists) [20]. Studies were not assigned an overall quality score; rather, the first two CASP questions determined eligibility (any 'no' led to exclusion), while responses to the remaining items were utilized as a quality assessment to determine methodological rigor and validity of the included studies [21]. The CASP Qualitative Studies Checklist asks ten questions to help make sense of qualitative research. These include statement of the aim, methodology, research design, recruitment, data collection, relationship between researcher and participants, ethical issues, data analysis, statement of findings, and the value of the research (CASP Checklists: https://casp-uk.net/casp-tools-checklists/). Each question was answered with the following categories: yes, can't tell, or no. Question ten was answered with an open-ended versus categorized response. Any discrepancies in the answers were discussed until consensus was reached.

Additionally, the GRADE-CERQual was used to assess the confidence of the findings. The GRADE-CERQual is an approach that helps to determine how much confidence to place in reviews of qualitative research. We followed the four components of this approach: [1] methodological limitations [2], coherence [3], adequacy of data, and [4] relevance [22].

Data extraction

The data extraction occurred with a minimum of two independent authors. Discrepancies in the data extraction process were discussed by the authors until consensus was reached, and finally checked by the first author (JRC). A data extraction form was developed including: publication details, author, year of publication, study name, aim, setting, dates of study, country of study, study design, population (recruitment strategy, demographics [including gender], sample size, and inclusion criteria), phenomena of interest (wishes, needs, and experience), and context (physical activity and exercise).

Data synthesis

Data from the articles were synthesized using a thematic approach, which organized and summarized findings from the included studies under key thematic headings [23]. The three-step process for thematic synthesis outlined by Thomas and Harden was used [24]. In step one, line-by-line coding was conducted by five independent authors (JRC, MB, CM, FB, CJW). In step two, codes were organized into related areas and descriptive themes were selected. Finally, in step three, analytical themes were developed. Both steps two and three were conducted by two authors (CB, CJW).

Results

Study selection

We identified 27,415 articles through the database and reference searches. After removal of 12,082 duplicates, 233 articles were screened in full text and 10 articles met all inclusion criteria for data extraction. The two most common reasons for exclusion were wrong study population or wrong phenomenon of interest. The sequence of the search process can be seen in Fig. 1.

Study characteristics

The study characteristics are shown in Table 1.

Design characteristics

Of the ten included studies, all but one had a qualitative study design (25–30, 32–34, and the remaining study had a mixed method design [31]. Five of the studies collected

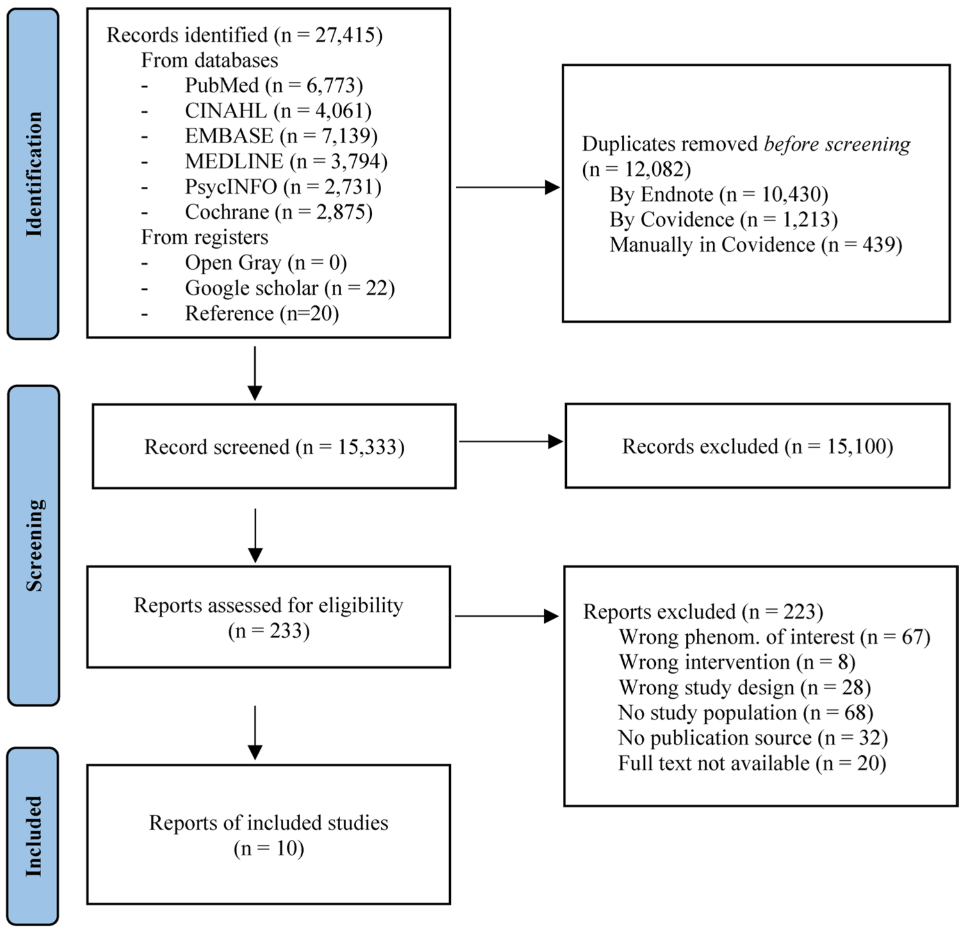

Fig. 1 Flowchart showing the search and selection of studies

data using individual interviews [25–27, 32], six studies used focus group interviews [29–34], and one study supplemented interviews with observations [25]. Two of the studies included more than one method to collect their data [25, 32].

Characteristics of location

The included studies were published between the years 1989–2022. Two studies were carried out in the United States [26, 30] and two in Canada [27, 28]. One study took place in each of the following countries: Barbados [25], Qatar [29], Singapore [31], England [32], Denmark [33] and New Zealand [34].

Participant characteristics

In total, 297 persons participated in the included studies, with a mean age of 40 years and ranging from 18 to 70 years. Their mean BMI were 35 (range 25–52), and 60% of the population were females. Two studies investigated only men [31, 34], and four looked only at women [25, 26, 28, 30].

Characteristics of phenomena of interest

The phenomena of interest encompassed barriers and facilitators [25, 28], experiences [26, 33], and perceptions [29, 30], reactions [26], behaviors [21], beliefs [22], perspectives and motivation [26], views and attitudes [27], and thoughts and expectations [34].

Characteristics of Context

Seven studies looked at physical activity labelled as “physical activity” [25, 27–30, 33, 34], while three studies labelled this construct as “exercise” [26, 31, 32].

Quality assessment

All the studies were assessed as having valid results according to the The CASP Qualitative Studies Checklist. All studies were also assessed to determine that they had an appropriate research design. Two of the studies did not have an appropriate recruitment strategy [26, 31]. In all the studies, data was collected in a way that addressed the research question. In six of the studies, the relationship between the researcher and the participants was not clearly described [27–32]. In two of the studies, there was not a clear description of how ethical issues were taken

Table 1 Description of the included studies

Reference (First author, year, country (number in reference list))	Population (Number, age, gender)	Study design	Phenomena of Interest: Wishes, needs, and experience	Context: Physical activity and exercise
Alvarado, Murphy, and Guell, 2015Barbados [25]	17 women, Age: 27–34 years	Qualitative case study (qualitative interviews and semi-structured participant observation)	Barriers and facilitators to physical activity	Physical activity
Bain et al. 1989 USA [26]	18 women Age: 25–61 years	Interviews	Reactions, experiences, behaviours	Exercise
Belanger-Gravel et al., 2013Canada [27]	30 men and Women (40%) Age: 62.7 ± 4.1 years	Theory of planned behaviour elicitation study (methods = interviews)	Beliefs	Physical activity
Dikareva et al., 2016Canada [28]	12 women Age: mean; 47.6 years	Semistructured interview	Barriers, facilitators, and motivators	Physical activity
Donnelly et al. 2018Qatar [29]	128 men and women Age: ≥ 18	Focus group interviews	Perceptions	Physical activity
Dunn, 2008USA [30]	14 women Age: 45–66 years	semistructured focus-group interviews	Perceptions	Physical activity
Gorny et al., 2021Singapore [31]	15 men Age: mean; 20.1 years	Focus group interviews	Perspectives and motivation	Exercise
Guess, 2012England [32]	25 women, 5 men Age: 21–65 years	Focus groups, individual interviews, and follow-up individual interviews	Views and attitudes	Exercise
Toft et al., 2022Denmark [33]	7 women Age: 30–57 years 8 men Age: 25–68 years	Focus group interview	Experiences	Physical activity
Warbrick, Wilson, and Boulton, 2016New Zealand [34]	18 men Age: 28–72 years	Focus group interview	Thoughts and expectations	Physical activity

into consideration [32, 34]. In four of the studies, the data analysis process was not sufficiently described [27, 31, 32, 34]. In one study, the statement of findings was unclear. The quality of the studies differed with five studies having had high quality [25, 28–30, 33], and five studies having had moderate quality [26, 27, 31, 32, 34]. See Table 2 for more detail about the critical assessment results.

Synthesis of results

In the thematic analysis, 283 codes were identified, which were then organized into 15 descriptive themes. We then identified two primary themes and seven subthemes.

Theme 1 - Coping

All the included studies reported how the participants' coping skills affected their ability to maintain or increase their physical activity.

Health

Most of the studies described health both as an important motivator and a barrier to being physically active. Dikareva et al. reported that limited mobility and joint pain, as well as depression and low mood, were major barriers to physical activity. A history of traumatic life events (e.g., childhood sexual abuse and divorce) were also shown to have a negative impact an individual's ability to be physically active [28]. Donnelly et al. reported that participants were well aware of the health benefits of physical activity. Motivation for reducing weight and preventing EBW and other chronic diseases were the most common reported facilitators for participants to engage in physical activity [29]. Guess described that the participants were aware of health benefits of physical activity. One participant in the study stated: "Yeah, I know for sure that people with good general fitness generally, tend to have a better state of mind, like less pressure" and another participant described that he thought that physical activity could work to combat the risk of diabetes and heart disease [32].

Knowledge and skills

Many of the studies described a lack of knowledge as another major barrier for being physically active. Bain et al. described that gaining knowledge could support regular exercise [26]. Participants in research by Guess shared that lack of knowledge was a barrier to physical activity with one participant stating, "I can't find an exercise that isn't painful - so that's where I would need specialist help" [32]. Many participants were not aware of exercise options that were affordable [20], and a lack of knowledge made it difficult for many to navigate physical activity [34]. Several of the studies reported that increasing skills

Table 2 Critical assessment results

Citation (first author, year, of publication, number in reference list)	Q1 Was there a clear statement of the aims of the research?	Q2 Is a qualitative methodology appropriate?	Q3 Was the research design appropriate to address the aims of the research?	Q4 Was the recruitment strategy appropriate to the aims of the research?	Q5 Was the data collected in a way that addressed the research issue?	Q6 Has the relationship between researcher and participants been adequately considered?	Q7 Have ethical issues been taken into consideration?	Q8 Was the data analysis sufficiently rigorous?	Q9 Is there a clear statement of findings?	Q10 How valuable is the research?	Score
Alvarado, Murphy, and Guell, 2015 [25]	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	10
Bain, Wilson, and Chaikind, 1989 [26]	Y	Y	Y	N	Y	Y	Y	Y	C	Y	8,5
Belanger-Gravel et al., 2013 [27]	Y	Y	Y	Y	Y	N	Y	N	Y	Y	8
Dikareva et al., 2016 [28]	Y	Y	Y	Y	Y	C	Y	Y	Y	Y	9,5
Donnelly et al., 2018 [29]	Y	Y	Y	Y	Y	C	Y	Y	Y	Y	9,5
Dunn, 2008 [30]	Y	Y	Y	Y	Y	C	Y	Y	Y	Y	9,5
Gorny et al., 2021 [31]	Y	Y	Y	N	Y	C	Y	C	Y	Y	8
Guess, 2012 [32]	Y	Y	Y	Y	Y	N	C	C	Y	C	7,5
Toft et al., 2022 [33]	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	10
Warbrick, Wilson, and Boulton, 2016 [34]	Y	Y	Y	Y	Y	Y	C	N	Y	Y	8,5

Y: Yes,
C: Can't Tell,
N: No

and knowledge could support being physically active. Growing awareness of physical activity benefits allowed participants to see that there were other exercise options that they found motivating and supportive [26, 32].

Adaptation to everyday life

Studies reported different kinds of adaptation to everyday life such as lack of time, work conflicts, school obligations, payment for exercise (e.g., fitness club abonnements and personal trainer) and integrating physical activity into daily routine. Dikareva et al. reported that "participants often struggled to make time for daily physical activity because of competing responsibilities that included occupational and educational demands, as well as family responsibilities" [28], and Dunn described that lack of time was a major reason for not walking [30]. Warbrick, Wilson, and Boulton reported that it was difficult to prioritize finances for the expenses to pay for a gym membership [34]. Several studies described that integration of physical activity into a daily routine was a facilitator to maintaining physical activity (e.g., active transportation, playing with kids or social walking). Another facilitator was joyfulness in the act of exercise [25, 27, 33].

Self-perception and motivation

Several studies reported that lack of self-efficacy, self-consciousness and concerns about physical appearance were barriers for physical activity engagement because the participants felt shame, embarrassment, and discomfort when engaging in physical activity [28, 31–33]. Toft et al. reported that one participant said: "I feel like a bull in a china shop. I feel really clumsy [doing yoga]. All those positions you have to get into. if I'm not able to do them, I get mad at myself and think "I'm too fat for this anyway"" [33]. Some of the studies described that the desire for weight loss was a reason for seeking an exercise program [26, 32]. Some women reported that the inability to fit into their clothes was a motivator to increase levels of physical activity [26, 28]. Guess reported that a lack of motivation was a barrier, with one participant sharing: "I find it hard to stick to [weight loss] targets because if I don't meet them I get real down so now I try not to make such rigid targets for myself. if it is not going to happen I am going to get de-motivated" [32].

Theme 2 - Environment

The second theme that emerged was environment, describing how the participants' environment affected their ability to maintain or increase their physical activity.

Culture and family support

Several studies reported on the importance on family support for engagement in physical activity. Alvarado, Murphy, and Guell reported that gender roles in the family limited womens' opportunities to exercise [25]. Dunn reported that family demands and prioritizing others' needs (e.g. friends and family members) were the most frequently reported reasons for not being physically active. The participants in this study also reported that a primary motivator for physical activity was that it also had some positive impact on the family [30]. Guess reported that one participant experienced a lack of family support for their exercise choices, and other participants described that they experienced too much encouragement or pressure from normal weight family members [32]. Donnelly et al. reported that some women experienced support from their husbands and family, while some families did not allow women to exercise. Women in this study reported that support from family and friends was a facilitator for physical activity [29]. Alvarado, Murphy, and Guell [25] and Donnelly et al. [29] reported on culture as a barrier. Specifically, Alvarado, Murphy, and Guell reported that the Barbadians described themselves as laid back people [25] and Donnelly et al. reported that in the culture in Qatar it is considered as a shame for women to exercise [29].

Peers and social support

Many studies reported on the importance on peers and social support of physical activity. Toft et al. reported that being physically active in a group was a new way of connecting with others, and that being active together with peers made people feel safe and a sense of belonging to a group [33]. Alvarado et al. described that several of the participants reported that they would be more active if they had a friend who would invite them to exercise, with participants reporting peer encouragement as a strong motivator [25]. In Warbrick et al. the participants referred to the "bros" around them in physical activity as the interaction they had with their male peers. The participants reported that it was easier to be physically active when they had their "bros" to motivate them and had peers to make them feel responsible. They also reported that a lack of peer support contributed to a loss of motivation [34]. Guess [32] and Belanger-Gravel et al. [27] reported that exercising alone caused increased boredom and added a barrier to physical activity [27, 32]. "Being alone while exercising is another important barrier among overweight/obese older adults. Building buddy systems, offering group sessions or directing individuals to community resources to favor social interaction could help participants maintain their participation in regular physical activity" [27].

Settings

Various studies underscored the importance of settings. Toft et al. described that the setting that the physical activity occurred in and the experiences connected with the setting were important. Different settings (e.g. places, environment and surroundings) provided different experiences of "feeling home" throughout the physical activity [33]. Toft et al. also reported that the participants had increased discomfort in places where they did not felt at home (e.g., in settings where they felt compared to others or judged) [33]. Alvarado, Murphy, and Guell reported that some women described that they were unable to perform exercises at home because they lived in spaces that were too small [25]. Donnelly et al. reported a barrier for physical activity being settings that were not accessible because of distance, fees etc [29].

Summary of Qualitative Findings

The confidence of the findings was assessed using "Confidence in the Evidence from Reviews of Qualitative Research" (GRADE-CERQual) [22]. The results are shown in a Summary of Qualitative Findings table (SoQF table) [22] in Table 3.

Discussion

This study aimed to explore how physical activity programs should be designed to effectively meet the wishes of individuals with EBW, identifying essential factors to consider when structuring such programs. While the included studies primarily addressed barriers, facilitators, and perceptions rather than directly examining how physical activity should be structured according to participants' preferences, these findings nevertheless provide important indirect insights, offering useful information for practice. Two overarching themes, coping and environment, emerged from the analysis, each encompassing specific barriers and facilitators that shape the ability of individuals with EBW to engage in physical activity.

One significant insight from this review was the role of health as both a barrier and a potential facilitator for physical activity. Poor physical and mental health were frequently cited as obstacles, where individuals felt limited in their ability to exercise despite understanding the potential health benefits. This aligns with findings from Baillot et al., which highlight physical discomfort and pain as common barriers in similar populations [35]. However, this dual role suggests that when individuals are guided to experience tangible health benefits from physical activity, health can transform into a powerful motivator [36]. Offering low-impact, accessible options might allow individuals to see gradual improvements, reframing health from a barrier into a pathway for engagement.

Additionally, a lack of knowledge and skills regarding physical activity also emerged as a critical barrier. Some participants expressed uncertainty about which activities were safe or feasible given their health and weight status. Others were unaware of options available in their communities. This knowledge gap can reduce confidence and motivation to participate in physical activity, a barrier noted in similar research on exercise adherence in people with limited physical activity experience [37]. Research by Malherbe et al. found that self-efficacy was a significant determinant of physical activity adherence, and that assessing an individual's knowledge and confidence surrounding exercise was an important consideration for longterm participation [38]. However, other research has also found that knowledge is a predictor of physical activity participation for older adults, but not for middle-aged and younger adults [39]. Therefore, the understanding gained from this theme alongside previous research indicates that providing accessible, practical education on physical activity participation, especially for older adults, could make physical activity feel more manageable and increase engagement among people with EBW.Adapting physical activity to everyday life was another important facilitator highlighted by participants. For many, competing commitments like work or school limited time available for physical activity, and the financial costs of gyms or trainers posed additional constraints [1]. Studies by Burgess et al. and Jessen-Winge et al. similarly observed that integrating physical activity into daily routines can be more sustainable [40, 41]. Incorporating activities within existing routines like active commuting (e.g., walking or cycling to work) or promoting fun, family-oriented exercises that don't feel like formal exercise might help people with EBW incorporate movement into their lives naturally [42]. Other interwoven activities that have been discussed in the literature to increase physical activity participation includes office exercise breaks, which is another option more efficiently integrate active time into an individual's daily schedule [43].

Self-perception and motivation emerged as complex barriers, with many participants reporting low self-esteem, shame, and discomfort with their physical appearance as obstacles. Participants expressed discomfort in settings where they felt exposed to others' gaze, with feelings of inadequacy affecting their willingness to engage in physical activity. Such psychological barriers are common in marginalized groups and have been documented by Caperchione et al., who noted similar issues of self-consciousness in women and older adults impacting physical activity participation [44]. Previous research has also indicated that communication of professionals working with individuals with EBW is particiluary impactful in addressing factors like motivation to improve participation [45]. Communication methods like motivational interviewing and fostering environments where individuals feel supported and unjudged could support motivation and self-consciousness of individuals, thereby supporting physical activity participation.

Table 3 CERQual summary of qualitative findings

Summary of review finding	Studies contributing to the review finding	CERQual assessment of confidence in the evidence	Explanation of CERQual assessment
Health	Bain, Wilson, and Chaikind [26] Alvarado, Murphy, and Guell [25] Guess [32] Dikareva et al. [28] Donnelly et al. [29] Belanger-Gravel et al. [27] Gorny et al. [31]	Moderate	It is graded as moderate confidence because there is a minor concern regarding methodological limitations, relevance, coherence, and adequacy
Knowledge and skills	Bain, Wilson, and Chaikind [26] Alvarado, Murphy, and Guell [25] Guess [32] Dikareva et al. [28] Donnelly et al.[29] Gorny et al.[31] Warbrick, Wilson, and Boulton [24]	Moderate	It is graded as moderate confidence because there is a minor concern regarding methodological limitations, relevance, coherence, and adequacy
Adaptation to everyday life	Bain, Wilson, and Chaikind [26] Alvarado, Murphy, and Guell [25] Donnelly et al. [29] Belanger-Gravel et al. [27] Warbrick, Wilson, and Boulton [34] Toft et al. [33] Dunn [30]	Moderate	It is graded as moderate confidence because there is a minor concern regarding methodological limitations, relevance, coherence, and adequacy
Self-perception and motivation	Bain, Wilson, and Chaikind [26] Alvarado, Murphy, and Guell [25] Belanger-Gravel et al. [27] Gorny et al. [31] Toft et al. [33] Guess [32] Dikareva et al. [28] Donnelly et al. [29] Dunn [30]	Moderate	It is graded as moderate confidence because there is a minor concern regarding methodological limitations, relevance, coherence, and adequacy
Culture and family support	Alvarado, Murphy, and Guell [25] Bain, Wilson, and Chaikind [26] Gorny et al. [31] Donnelly et al. [29] Guess [32] Belanger-Gravel et al. [27] Dunn [30] Dikareva et al. [28]	Moderate	It is graded as moderate confidence because there is a minor concern regarding methodological limitations, relevance, coherence, and adequacy
Peers and social aspects	Warbrick, Wilson, and Boulton [34] Alvarado, Murphy, and Guell [25] Belanger-Gravel et al. [27] Guess [32] Toft et al. [33] Bain, Wilson, and Chaikind [26] Dunn [30] Dikareva et al. [28]	Moderate	It is graded as moderate confidence because there is a minor concern regarding methodological limitations, relevance, coherence, and adequacy
Settings	Alvarado, Murphy, and Guell [25] Bain, Wilson, and Chaikind [26] Belanger-Gravel et al. [27] Guess [32] Donnelly et al. [29] Dikareva et al. [28] Toft et al. [33]	Moderate	It is graded as moderate confidence because there is a minor concern regarding methodological limitations, relevance, coherence, and adequacy

The environment in which participants attempted to exercise also played a significant role. Family support and cultural expectations shaped participants' willingness and ability to engage in physical activity. For example, some participants described cultural norms as limiting, especially where gender roles or other family obligations conflicted with their physical activity goals. This aligns with previous findings where societal expectations can restrict physical activity in certain populations [40, 44, 46]. Involving family members in awareness programs, fostering a more supportive culture around physical activity, and offering culturally inclusive exercise options could help mitigate these environmental barriers and foster supportive networks.

Social support from peers was also found to be an important facilitator. Participants often cited that having peers who encouraged them or joined them in activities provided a sense of security and increased their motivation. Similar findings have been reported elsewehere, where peer support has been shown to both motivate and sustain physical activity in people with EBW [35]. However, studies have also found that social support can be perceived both as a motivation and a barrier [35, 40, 41]. Burgess, Hassmén, and Pumpa reported that peers can have negative influence, which are different from the findings in this review [40]. Designing positive social supports through opportunities like structured group sessions or buddy systems could capitalize on this benefit, creating a supportive, social environment that meets the need for companionship while reducing feelings of isolation. Programs that promote peer involvement may also help participants feel that their efforts are validated and shared by others in a similar situation, further reinforcing commitment to physical activity.

Finally, the physical setting where activity took place influenced participants' comfort and motivation. For some, typical gym settings felt intimidating or inappropriate, particularly when exercising among people without EBW. Issues like space limitations at home also constrained activity, echoing findings from Burgess et al., who identified setting as a potential barrier to physical activity [40]. Offering alternative options, such as virtual exercise classes or community programming, may make physical activity more accessible, providing spaces where participants feel comfortable and supported. For example, designing virtual exercise classes would allow individuals the opportunity to engage in physical activity programming from a space that feels comfortable to them. Additionally, developing community-based physical activity programs offers opportunities to engage diverse populations in interventions within more approachable settings like parks, community centers, schools, and libraries.

Limitations

Throughout this systematic review, many efforts were made to identify all relevant research articles. However, it is possible that existing relevant research may have been overlooked due to publications being in different languages and being inaccessible. Moreover, the search strategy did not include terms related to program structuring, which may have limited the inclusion of studies on how physical activity can be planned and structured. Another limitation of this review was that the studies were from all over the world, taking place in different cultures and settings. Therefore, the different contexts that this research took place in could have affected the overall synthesis. There is also the potential of publication bias impacting the results of this systematic review, as researchers may be less linkely to submit negative or inconsequential results, skewing the synthesis of this review [47]. Additionally, the nature of this systematic review limits the ability to generalize the findings of the included studies to other populations [48]. The final limitation that needs to be noted relates to the quality of included articles. There were limitations in the quality of included studies as discussed previously limiting the conclusions that can be drawn from this systematic review.

Conclusions

This qualitative systematic review identified 10 articles. While none of the found articles explicitly investigated the wishes and needs of people with excess body weight to increase their level of physical activity, the studies did cover other perspectives of people with EBW regarding physical activity which add to knowledge supporting the planning and organization of physical activity for this population. These perspectives were synthesised into two primary themes (coping and environment) and seven subthemes (health, knowledge and skills, adaptation to everyday life, self-perception and motivation, culture and family support, peers and social aspects and settings). This review demonstrated how various factors related to personal wishes could be implemented in physical activity programming to improve participation for individuals with EBW. Addressing health barriers, improving access to knowledge, integrating activities into daily life, and offering supportive, inclusive environments are essential strategies for increasing engagement. Future research should explore more comprehensive aspects of personal wishes related to physical activity. Multiple factors related to preferences serve to influence behavioral participation. While the research systhesized in this review explored some aspects of personal preferences related to physical activity for individuals with EBW, there are still other wishes that should be explored to better understand how to design programming for this population. Preferential factors of individuals with EBW regarding exercise that were not explored in any of the included articles that would serve to improve physical activity programming include the followining: when individuals prefer to participate in exercise, who individuals want to partake in exercise with, and how individuals desire to engage in exercise. Exploring these wishes of individuals with EBW will more appropriately guide physical activity design for this population.

Abbreviations
EBW Excess body weight

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Authors' contributions
JRC and CB were responsible for the overall planning and design of the study, the development of the PICo questions, and the drafting of the initial article. CB and SSN conducted the preliminary search, while JRC, CB, and SSM undertook the final search, guided by the PICo framework. The initial screening of records for relevance based on title and abstract was performed by JRC and CB. The subsequent full-text review of selected articles was carried out by CM, FB, HB, SSH, JKK, CJB, SSN, and CJW, who assessed eligibility using a predefined selection form with clear inclusion criteria. All ten authors contributed to the data extraction process and to the writing of the manuscript, and all authors reviewed and approved the final version of the paper.

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